Property-VI

(i) $\sin^{-1}x+\sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$, where either $x^2+y^2\le 1$ or $xy<0$.

(ii) $\sin^{-1}x-\sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2}-y\sqrt{1-x^2}\right)$, where either $x^2+y^2\le 1$ or $xy>0$.

(iii) $\cos^{-1}x+\cos^{-1}y=\cos^{-1}\left[xy-\sqrt{1-x^2}\sqrt{1-y^2}\right]$, if $x+y\ge 0$.

(iv) $\cos^{-1}x-\cos^{-1}y=\cos^{-1}\left[xy+\sqrt{1-x^2}\sqrt{1-y^2}\right]$, if $x\le y$.

(v) $\tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\dfrac{x+y}{1-xy}\right)$, if $xy<1$.

(vi) $\tan^{-1}x-\tan^{-1}y=\tan^{-1}\left(\dfrac{x-y}{1+xy}\right)$ if $xy>-1$.

Proof

(i) Let $A=\sin^{-1}x$. Then, $x=\sin A$; $A\in\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$; $|x|\le 1$ and $\cos A$ is positive

Let $B=\sin^{-1}y$. Then, $y=\sin B$; $B\in\left[-\dfrac{\pi}{2},\dfrac{\pi}{2}\right]$; $|y|\le 1$ and $\cos B$ is positive

Now, $\cos A=+\sqrt{1-\sin^2 A}=\sqrt{1-x^2}$ and $\cos B=+\sqrt{1-\sin^2 B}=\sqrt{1-y^2}$

Thus, $\sin(A+B)=\sin A\cos B+\cos A\sin B$

$= x\sqrt{1-y^2}+y\sqrt{1-x^2}$, where $|x|\le 1$; $|y|\le 1$ and hence, $x^2+y^2\le 1$

Therefore, $A+B=\sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$

Thus, $\sin^{-1}x+\sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2}+y\sqrt{1-x^2}\right)$, where either $x^2+y^2\le 1$ or $xy<0$.

Similarly, other results are proved.

Property-VII

(i) $2\tan^{-1}x=\tan^{-1}\left(\dfrac{2x}{1-x^2}\right), |x|<1$ (ii) $2\tan^{-1}x=\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right),\ x\ge 0$

(iii) $2\tan^{-1}x=\sin^{-1}\left(\dfrac{2x}{1+x^2}\right), |x|\le 1$.

Proof

(i) By taking $y=x$ in Property-VI (v) , we get the desired result

$2\tan^{-1}x=\tan^{-1}\left(\dfrac{2x}{1-x^2}\right), |x|<1$.

(ii) Let $\theta=2\tan^{-1}x$. Then, $\tan\dfrac{\theta}{2}=x$.

The identity $\cos\theta=\dfrac{1-\tan^2\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}}=\dfrac{1-x^2}{1+x^2}$ gives $\theta=\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$.

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